Project: Worked examples
Title: Thick-walled cylinder stresses (Lamé)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Radial, hoop and axial stresses at the bore and the outer surface of a thick-walled cylinder under internal and external pressure, the von Mises and Tresca equivalents, utilisation against an allowable, and the thin-wall estimate for comparison.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Lamé constants from the bore and outside pressures | $A = (a^2 p_i - b^2 p_o)/(b^2 - a^2)$; $B = a^2 b^2 (p_i - p_o)/(b^2 - a^2)$ |
| Radial and hoop stress at radius r | $\sigma_r = A - B/r^2$; $\sigma_\theta = A + B/r^2$ |
| Axial stress, closed or open ends | $\sigma_z = A$ (closed); $\sigma_z = 0$ (open) |
| Equivalent stresses from the three principal stresses | von Mises; Tresca = the largest principal difference |

Basis: Thick-walled cylinders[1](Ch. 13, Table 13.5); Stresses in pressurised cylinders[2](§3-14).

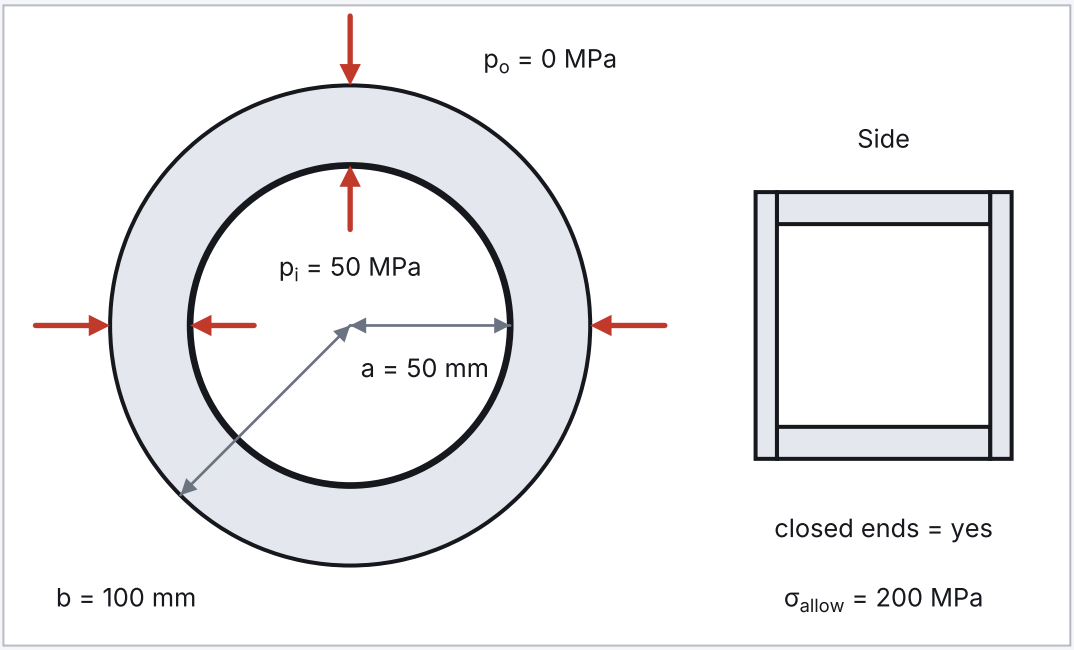



Figure 1 — A thick-walled cylinder in section, bore radius a, outer radius b, pressure inside and out; closed ends carry the pressures axially. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| a | Inner radius | 50 | mm | drawing: DRG-CYL-01 rev A | 0.1 … 5000 mm |
| b | Outer radius | 100 | mm | drawing: DRG-CYL-01 rev A | 0.1 … 5000 mm |
| $p_i$ | Internal pressure | 50 | MPa | spec: SPEC-PV-2 §3 design pressure | 0 … 1000 MPa |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $p_o$ | External pressure | 0 | MPa | assumption: Atmospheric outside. | 0 … 1000 MPa |
| closed | Closed ends (axial stress carried by the wall) | yes | | drawing: DRG-CYL-01 end caps rev A | any |
| E | Young's modulus (the elastic collapse pressure under external pressure only) | 207 | GPa | default | 1 … 500 GPa |
| v | Poisson's ratio (the elastic collapse pressure under external pressure only) | 0.3 | | default | 0.1 … 0.49 |
| $\sigma_{allow}$ | Allowable stress | 200 | MPa | spec: SPEC-PV-2 §5 | 1 … 3000 MPa |

## 3. Assumptions

A1. Long cylinder, elastic, isotropic; stresses vary with radius only. (Lamé solution; end effects neglected.)
A2. Closed ends carry the pressure end load as a uniform axial stress. ($\sigma_z$ = A.)
A3. $p_o$ (External pressure) is taken as 0 MPa. (Atmospheric outside.)

## 4. Working

[9] A = (a^2 * $p_i$ - b^2 * $p_o$) / (b^2 - a^2) [1](Table 13.5)
= ((50 mm)^2 × 50 MPa − (100 mm)^2 × 0 MPa) / ((100 mm)^2 − (50 mm)^2)
= 16.667 MPa
Lamé constant A

[10] B = a^2 * b^2 * ($p_i$ - $p_o$) / (b^2 - a^2) [1](Table 13.5)
= (50 mm)^2 × (100 mm)^2 × (50 MPa − 0 MPa) / ((100 mm)^2 − (50 mm)^2)
= 166670 MPa·mm²
Lamé constant B

[11] $\sigma_{r,a}$ = A - B / a^2
= 16.667 MPa − 166670 MPa·mm² / (50 mm)^2
= -50 MPa
Radial stress at the bore

[12] $\sigma_{t,a}$ = A + B / a^2
= 16.667 MPa + 166670 MPa·mm² / (50 mm)^2
= 83.333 MPa
Hoop stress at the bore

[13] $\sigma_{r,b}$ = A - B / b^2
= 16.667 MPa − 166670 MPa·mm² / (100 mm)^2
= 1.8626e-15 MPa
Radial stress at the outer surface

[14] $\sigma_{t,b}$ = A + B / b^2
= 16.667 MPa + 166670 MPa·mm² / (100 mm)^2
= 33.333 MPa
Hoop stress at the outer surface

[15] $\sigma_z$ = A
= 16.667 MPa
= 16.667 MPa
Axial stress (closed ends)

[16] $\sigma_{vm,a}$ = sqrt((($\sigma_{t,a}$ - $\sigma_{r,a}$)^2 + ($\sigma_{r,a}$ - $\sigma_z$)^2 + ($\sigma_z$ - $\sigma_{t,a}$)^2) / 2)
= sqrt(((83.333 MPa − (-50 MPa))^2 + ((-50 MPa) − 16.667 MPa)^2 + (16.667 MPa − 83.333 MPa)^2

= 115.47 MPa

von Mises stress at the bore

[17] $\sigma_{vm,b} = \text{sqrt}(((\sigma_{t,b} - \sigma_{r,b})^2 + (\sigma_{r,b} - \sigma_z)^2 + (\sigma_z - \sigma_{t,b})^2) / 2)$

= sqrt(((33.333 MPa - 1.8626e-15 MPa)^2 + (1.8626e-15 MPa - 16.667 MPa)^2 + (16.667 MPa - 33.333 MPa)^2) / 2)

= 28.868 MPa

von Mises stress at the outer surface

[18] $\sigma_{tr,a} = \max(\sigma_{t,a}, \sigma_{r,a}, \sigma_z) - \min(\sigma_{t,a}, \sigma_{r,a}, \sigma_z)$

= max(83.333 MPa, -50 MPa, 16.667 MPa) - min(83.333 MPa, -50 MPa, 16.667 MPa)

= 133.33 MPa

Tresca stress at the bore

[19] $\sigma_{tr,b} = \max(\sigma_{t,b}, \sigma_{r,b}, \sigma_z) - \min(\sigma_{t,b}, \sigma_{r,b}, \sigma_z)$

= max(33.333 MPa, 1.8626e-15 MPa, 16.667 MPa) - min(33.333 MPa, 1.8626e-15 MPa, 16.667 MPa)

= 33.333 MPa

Tresca stress at the outer surface

[20] $\sigma_{vm,max} = \max(\sigma_{vm,a}, \sigma_{vm,b})$

= max(115.47 MPa, 28.868 MPa)

= 115.47 MPa

Maximum von Mises stress

[21] $u = \sigma_{vm,max} / \sigma_{allow}$

= 115.47 MPa / 200 MPa

= 0.57735

Utilisation

[22] $\sigma_{thin} = (p_i - p_o) * (a + b) / 2 / (b - a)$

= (50 MPa - 0 MPa) × (50 mm + 100 mm) / 2 / (100 mm - 50 mm)

= 75 MPa

Thin-wall hoop estimate at the mean radius

[23] $k_{wall} = b / a$

= 100 mm / 50 mm

= 2

Wall ratio b/a

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $\sigma_{r,a}$ | -50 MPa | Radial stress, bore |
| $\sigma_{t,a}$ | 83.333 MPa | Hoop stress, bore |
| $\sigma_{r,b}$ | 1.8626e-15 MPa | Radial stress, outer |
| $\sigma_{t,b}$ | 33.333 MPa | Hoop stress, outer |
| $\sigma_z$ | 16.667 MPa | Axial stress |
| $\sigma_{vm,a}$ | 115.47 MPa | von Mises, bore |
| $\sigma_{vm,b}$ | 28.868 MPa | von Mises, outer |
| $\sigma_{tr,a}$ | 133.33 MPa | Tresca, bore |
| $\sigma_{tr,b}$ | 33.333 MPa | Tresca, outer |
| $\sigma_{vm,max}$ | 115.47 MPa | Maximum von Mises |
| u | 0.57735 | Utilisation |
| $\sigma_{thin}$ | 75 MPa | Thin-wall hoop estimate |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $k_{wall}$ | 2 | Wall ratio |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | von Mises against allowable | u = 0.57735 | u ≤ 1 | 0.577 | |

## 7. Applicability

WARNING — input.E: E (Young's modulus (the elastic collapse pressure under external pressure only)) was not entered; the template default 207 GPa is used.
WARNING — input.nu: ν (Poisson's ratio (the elastic collapse pressure under external pressure only)) was not entered; the template default 0.3 is used.

## 8. Verification

Template mech.thick-cylinder@2.1.0 Content hash 97bc46244f652b46 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.